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Decentralized Robust Connectivity Control in Flocking of Multi-robot Systems

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ABSTRACT In this paper, a global connectivity control method for decentralized multi-robot systems is proposed. This method can achieve decentralized connectivity control of multi-robot network under disturbances, which has no effect on the objective flocking control. Based on the gradient between the connectivity and robot positions, the proposed connectivity control method can make each robot move along the desired gradient direction, so as to achieve the control of the global connectivity. Indeed, the flocking method based on the potential of attraction and repulsion can ensure that the distance between robots is stable within the desired range. Then the security of flocking and the stability of communication are guaranteed. It is proved in this paper that both connectivity and objective flocking control have no effect on the stability of each other, under the condition that both controllers are bounded. Therefore, the global connectivity and configuration of the system can achieve the desired states. In addition, a robust control method based on integral sliding mode is designed in this paper, which can counteract the external disturbance and ensure the ideal dynamics of multi-robot systems. Finally, several numerical simulations are given to validate the effectiveness of the proposed control methods.

INDEX TERMS Multi-robot systems, distributed estimation, global connectivity control, potential energy function, robust control.

I. INTRODUCTION

Multi-robot systems (MRSs) have attracted great attention from researchers over the last three decades. The system composed of multiple robots can accomplish complex tasks in cooperation, which is difficult for a single robot. Indeed, MRSs have a wide range of applications in different fields, such as environmental mapping[1], cooperative search[2], UAV formation[3], and *et al*. The realization of cooperative tasks of MRSs depends on stable communication networks, that is, robots need to carry out the necessary information interaction to improve the capability of task execution. The interactive information includes but is not limited to the states and sensor data of each robot. Therefore, it is an important issue to preserve the connectivity of the communication network.

The connectivity control of MRS networks has also been widely studied in the last decade. The connectivity graph is often used to model the topology of the communication network[4]， while the connectivity control strategy is to ensure that the graph is connected over time. In addition, for the efficient use of communication resources, the control strategy is supposed to adjust the connectivity strength of the network according to tasks.

The premise of connectivity control is the measurement of global connectivity. As it is known from algebraic graph theory, the algebraic connectivity, defined as the second smallest eigenvalue of the Laplacian matrix, can be used to measure the connectivity of the network[5]. Meanwhile, the network is connected if and only if the algebraic connectivity is positive, and the greater the algebraic connectivity, the stronger the connectivity of the MRS network. However, robots cannot obtain global connectivity in a large system or a harsh environment. Each robot was able to maintain the local connectivity with limited information in [6] and [7]. The main idea of local connectivity maintenance is to design a controller to ensure

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that if a communication link is active initially, it will be always active. The advantage of this control algorithm is that it only needs local information and the maintenance of the connectivity is formally proven. However, it is often too restrictive to maintain each communication link. In practice, for the information exchange in MRSs, it is only necessary to guarantee the connectedness of the communication graph. The unnecessary links will consume limited communication resources of robots. In addition, the communication topology is not allowed to be changed in this method, which limits the flexibility of MRSs.

In order to solve this problem, more attention is paid to the global connectivity in practical applications. The global connectivity strategy allows the addition and deletion of links while the global connectivity is preserved over time, which is an important consideration in some task scenarios. A distributed connectivity estimation method based on the energy calculation of the connection matrix was proposed in [8]. The estimation method could give the upper and lower bounds of the global algebraic connectivity in each iteration, which could be used for global connectivity control. A method based on the distributed estimation of the spectrum of the Laplacian matrix was proposed in [9], where the global algebraic connectivity with small error could be obtained in finite time. A global connectivity estimation method based on standard power iteration is proposed in [10]. This method could estimate the second small eigenvalue of the Laplacian matrix and its corresponding eigenvector in a distributed manner, so that the global algebraic connectivity was obtained. Furthermore, the connectivity estimation of power iteration was improved in [11] to ensure the boundedness of estimation error.

With the results of the decentralized global connectivity estimation, a gradient-based control method was designed in [11] and [12], which could maintain the global connectivity in a distributed manner. A distributed method for global connectivity maintenance with unicycle kinematics was investigated in [13]. The proposed method guaranteed the global connectivity of systems under the nonholonomic constraint. As for directed connectivity graphs, connectivity estimation and control methods were proposed in [14]. The methods could maintain the connectivity of directed networks and increase the communication radius if necessary. A super-gradient method was used in conjunction with the decentralized algorithm in [15] to maximize the algebraic connectivity of the connection graph. Then, a potential-based control law was utilized to achieve the desired connectivity. For the network with constrained discrete-time linear dynamics, a distributed constrained connectivity control algorithm was proposed in [16]. In order to ensure the connectivity of the graph, the method enforced that there always existed a specific spanning tree. A decentralized method for the estimation and control of the connectivity with a random topology was proposed in [17], which could maximize the connectivity with the existence of realistic medium access control (MAC). A topology configuration method for the network of heterogeneous optical satellites was addressed in [18], where the problem of solving polynomials was relaxed to a convex optimization problem to maximize the weighted algebraic connectivity.

However, it is not always the best choice to enhance the global connectivity for MRSs. For example, the reduction of global connectivity can increase the coverage of MRSs in search or exploration tasks. As for UAV formation attacks, strong connectivity means more communication links, which increases the risk of detection by the enemy. In addition, strong connectivity will take up more unnecessary network resources, which is not conducive to the execution of overall tasks. To solve this problem, a distributed control algorithm is proposed in [19] to maintain, enhance, and reduce the global connectivity online, according to different stages of tasks.

In practical applications, MRSs often perform path or configuration control in addition to connectivity control. Therefore, the mutual influence of the two control strategies is also an important issue worthy of analysis. In the optimization problem of path planning, connectivity constraints were considered in [20], which provided a bounded control input for the robots to preserve the global connectivity. Connectivity constraints on the motion of MRSs were also addressed in [21], where authors provided a method to enforce the communication links among cooperative robots. A class of generalized potentials including connectivity discontinuities caused by unexpected obstacles or noises were designed in [22]. The new distributed protocol was utilized to preserve the global connectivity and to realize the optimization of the quadratic cost function. In addition, bounded control strategies were addressed in [23]-[25], where the global connectivity could be guaranteed over time and the desired configurations with collision avoidance were achieved.

However, the above studies only focus on the maintenance of initial communication links, which reduce the flexibility of the topology. Therefore, a distributed hybrid scheme composed of link switch and potential function was proposed in [26], which could retain or reject links to satisfy constraints of dynamic and connectivity. A distributed control strategy was investigated in [27] to maximize area coverage of MRS while ensuring reliable communication between robots. The proposed method could achieve the optimization of the area coverage and connectivity by alternating between the two objectives. The algorithm to maintain, increase, and control connectivity in MRS networks was discussed in [28], where convex optimization and sub-gradient descent algorithms were utilized to maximize the algebraic connectivity of networks. Meanwhile, applications of connectivity control to MRSs rendezvous, flocking, and formation control were also

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discussed in the paper. The connectivity maintenance problem for the MRSs with a bounded collective control objective was investigated in [29], where the proposed methodology could preserve the global connectivity with desired configurations. Unfortunately, the above control strategies can only maintain or increase the connectivity, and cannot adjust the connectivity according to the scenario flexibly. In addition, the MRSs are often used in a complex environment, so that the effect of unknown disturbances on the system stability is a problem worthy of study.

Considering the limitations of the current studies, the objective of this paper is to propose a decentralized control methodology to adjust the global connectivity, and analyze the interaction between the connectivity control term and the objective flocking control one. To summarize, the following contributions are made in this paper:

1) A decentralized control method to adjust the global connectivity of the MRS according to different tasks, which can change rather than just preserve the topology of the system.

2) A theoretical analysis to evaluate the interaction between the connectivity control term and the objective potential-based flocking controller, and to improve the universality of the proposed method.

3) A theoretical analysis of robustness of the proposed connectivity control method against the errors of the global connectivity estimation and a sliding mode method is utilized to ensure the stability of MRSs.

The rest of the paper is organized as follows. Section II contains a brief introduction to the graph theory and the estimation of algebraic connectivity. In section III, we propose the decentralized connectivity control method and analyze its effect on the objective potential-based flocking controller. In Section IV, we evaluate the robustness of the proposed connectivity control method on estimation errors, and a robust controller based on integral sliding mode is utilized to counteract the disturbance. Section V gives several simulations for validation purposes. Finally, the conclusion and future work are provided in section VI.

II. PRELIMINARIES

A. GRAPH THEORY AND GLOBAL CONNECTIVITY

Given a system composed of N robots moving in a d-dimensional space. Let us denote $p_i \in \mathbb{R}^d$ as the position of robot i. There is no central node in the network, and robots exchange information on an undirected network with the communication radius R. Let $\mathcal{N}$ represent the set of all robots in the system, and $E \subseteq \mathcal{N} \times \mathcal{N}$ represent the set of communication links. For robot i, we denote its set of communication neighbors as $\mathcal{N}_i$. That is, if $\|p_i - p_j\| \le R$, then $j \in \mathcal{N}_i$. Besides, we denote $e_{ij} = 1$ when $j \in \mathcal{N}_i$, where $e_{ij} \in E$ is the edge between robot i and j, and $e_{ij} = 0$ otherwise. Thus, the global connectivity graph can be described as

$$G = \{\mathcal{N}, E\} \tag{1}$$

The connectivity of the system is described by the adjacency matrix $A \in \mathbb{R}^{N \times N}$, where $a_{ij} \in A$ represents the weight of the edge e_{ij}, and $a_{ij} \ge 0$. Let $a_{ij} = a_{ji}$, since the graph G is undirected. The weighted adjacency matrix is defined as

$$a_{ij} = \begin{cases} e^{\left(R^2 - \|p_i - p_j\|^2\right)/2\theta^2} - 1 & if \;\; j \in \mathcal{N}_i \\ 0 & elsewise \end{cases} \tag{2}$$

where $\|p_i - p_j\|$ is the adjacency distance between robot i and j, θ is a tuning parameter. In this paper, we define $\theta = \sqrt{R^2/(2ln2)}$ to ensure $a_{ij} \in (0,1)$. The degree matrix of graph G is $D = diag(\{d_i\})$, where $d_i = \sum_{j=1}^{N} a_{ij}$ represents the degree of the robot i in graph G.

The weighted Laplacian matrix of the graph G is defined as[19]

$$L = D - A \tag{3}$$

It is obvious that $L\boldsymbol{1} = 0$, where $\boldsymbol{1}$ is a column vector of all ones. Given the eigenvalues of L are real and satisfy $0 = \lambda_1 \le \lambda_2 \le \cdots \le \lambda_N$. It is known that $\lambda_2 > 0$ if and only if the graph G is connected. Therefore, λ_2 is defined as the algebraic connectivity to measure the connectivity of a graph, and the larger the λ_2 is, the stronger the graph connectivity is.

B. GLOBAL CONNECTIVITY ESTIMATION

The global connectivity of the system is time-varying with the movement of robots. Therefore, robots are supposed to estimate the algebraic connectivity online in order to achieve the control of global connectivity.

A power iteration method for the estimation of global connectivity was investigated in [10], where the corresponding eigenvector v_2 is estimated ($Lv_2 = L\lambda_2$), which is then used to determine the algebraic connectivity λ_2. The estimation algorithm of v_2 is

$$\dot{\tilde{v}}_2 = -k_1 \mathrm{Ave}\left(\left\{\tilde{v}_2^i\right\}\right)\boldsymbol{1} - k_2 L\tilde{v}_2 - k_3\left(\mathrm{Ave}\left(\left\{\left(\tilde{v}_2^i\right)^2\right\}\right) - 1\right)\tilde{v}_2 \tag{4}$$

where $\tilde{v}_2 = [\tilde{v}_2^1, \cdots, \tilde{v}_2^N]$ represents the estimation result of v_2, $k_1, k_2, k_3 > 0$ represent the control gains, and the function $\mathrm{Ave}(\{x^i\}) = \left(\sum_{i=1}^{n} x^i\right)/n$ is to average the elements of a vector. As is proven in [10], the sufficient and necessary conditions of $\lim_{t\to\infty} \tilde{v}_2 = v_2$ are $k_1 > k_2\lambda_2$ and $k_3 > k_2\lambda_2$, and the result satisfies $\|\tilde{v}_2\|^2 = N(k_3 - k_2\lambda_2)/k_3$. Therefore, the estimation of global connectivity is

$$\tilde{\lambda}_2 = \frac{k_3}{k_2}\left(1 - \frac{\|\tilde{v}_2\|^2}{N}\right) = \frac{k_3}{k_2}\left(1 - \mathrm{Ave}\left(\left\{\left(\tilde{v}_2^i\right)\right\}\right)\right) \tag{5}$$

where the value of k_1, k_2, k_3 can be guaranteed by the maximum of λ_2 which is calculated by the weighted Laplacian matrix when $a_{ij} = 1$ for all robots.

Notably, the estimation algorithm in (5) is centralized. But in practice, each robot cannot get the entire L and v_2, so that a distributed method needs to be designed. The function $\mathrm{Ave}(\cdot)$ in the centralized method can be replaced by the distributed PI average consensus estimator in [30]

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$$\begin{cases} \dot{z}^i = \gamma\left(\alpha^i - z^i\right) - \eta_1 \sum_{j\in\mathcal{N}_i}\left(z^i - z^j\right) + \eta_2 \sum_{j\in\mathcal{N}_i}\left(\omega^i - \omega^j\right) \\ \dot{\omega}^i = -\eta_2 \sum_{j\in\mathcal{N}_i}\left(z^i - z^j\right) \end{cases} \tag{6}$$

where α^i is the input of each robot, z^i is the average estimation of $\sum_{i=1}^{N}\alpha^i/N$, $\gamma > 0$ is the update rate, and η_1, η_2 are estimator gains. The convergence of the estimator has been proved in [30], and the estimation error convergences to a ball around zero whose size is related to the change rate of the input, while constant input producing the error converging to zero [30]. Therefore, the error can be written as $e_i = \sum_{j\in\mathcal{N}_i}\alpha^j/N_i - \sum_{j=1}^{N}\alpha^j/N$.

It is known from (4) that the robot i runs two PI estimators to perform the distributed estimation. Let us define the first one to estimate $\mathrm{Ave}(\{\tilde{v}_2^i\})$ with input $\alpha_1^i = \tilde{v}_2^i$ and output z_1^i, and the second one to estimate $\mathrm{Ave}(\{(\tilde{v}_2^i)^2\})$ with the input $\alpha_2^i = (\tilde{v}_2^i)^2$ and output z_2^i. Thus, the distributed estimation method can be rewritten to

$$\dot{\tilde{v}}_2^i = -k_1 z_1^i - k_2 \sum_{j\in\mathcal{N}_i} a_{ij}\left(\tilde{v}_2^i - \tilde{v}_2^j\right) - k_3\left(z_2^i - 1\right)\tilde{v}_2^i - k_4\left|\tilde{v}_2^i\right|\tilde{v}_2^i \tag{7}$$

where the additional term $k_4\left|\tilde{v}_2^i\right|\tilde{v}_2^i$ is necessary to ensure the boundedness of the estimation error of λ_2 [11].

The distributed estimation z_2^i is utilized to replace the $\mathrm{Ave}(\{(\tilde{v}_2^i)^2\})$ in (5), then the global connectivity computed in robot i is

$$\tilde{\lambda}_2^i = \frac{k_3}{k_2}\left(1 - z_2^i\right) \tag{8}$$

It is known from (8) that the estimation error of λ_2 comes from the average consensus estimator result z_2^i. Since z_2^i is proved to be bounded in [11] and [30], there exits $\exists \varDelta > 0$, then

$$\left|\tilde{\lambda}_2^i - \lambda_2\right| \le \varDelta \tag{9}$$

where $\varDelta = k_3 e_i / k_2$. That is, the connectivity estimation error of each robot is bounded.

III. DECENTRALIZED CONNECTIVITY CONTROL

Consider the d-dimensional space composed of N moving robots, the dynamics of each robot i is described as

$$\dot{p}_i = f_i(\boldsymbol{p}) + u_i \tag{10}$$

where $f_i(\boldsymbol{p}): \mathbb{R}^{Nd} \to \mathbb{R}^d$ represents the control term related to robots flocking, $\boldsymbol{p} = [p_1^T, p_2^T, \cdots, p_N^T]^T$ is the stacked vector of per-robot state with $p_i \in \mathbb{R}^d$, and $u_i \in \mathbb{R}^d$ is the connectivity control of each robot.

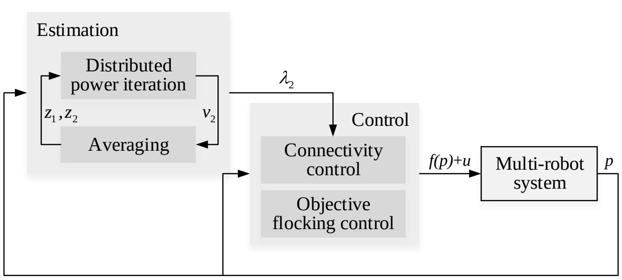

FIGURE 1. Proposed architecture of global connectivity control strategy with the objective flocking control

Although the dynamics in (10) is a very simplified model, it can be extended to a higher-order dynamical system with the method described in [31], where the Lagrangian dynamical system is considered. Therefore, the proposed control method based on first-order dynamics in this paper can still be effectively utilized to control real MRSs.

Assumption 1: Considering the physical limitation of the actuator capabilities in practice, there exist $U_0 > 0 \in \mathbb{R}$ and $U_c > 0 \in \mathbb{R}$ in the dynamics of the given MRS such that

$$\begin{gathered} \left\| f_i(\boldsymbol{p}) \right\| \le U_0, \ \forall i \in [1, 2, \cdots, N] \\ \left\| u_i \right\| \le U_c, \ \forall i \in [1, 2, \cdots, N] \end{gathered} \tag{11}$$

where U_0 and U_c which mean the upper bounds of the controllers and $U_0 + U_c$ represents the maximum output of the actuator.

The proposed architecture is illustrated in Fig.1. Specifically, the global connectivity control strategy relies on the result of distributed estimation and cooperates with the objective flocking term to achieve the control of both the global connectivity and the desired configuration of the MRS.

A. ANALYSIS OF CONNECTIVITY CONTROL

The connection states of robots can be described as the weighted adjacency matrix in (2). Then, the gradient of global connectivity λ_2 with respect to p_i is

$$\begin{aligned} \nabla_{p_i}\lambda_2(\boldsymbol{p}) &= v_2^T \frac{\partial L}{\partial p_i} v_2 \\ &= \sum_{j\in\mathcal{N}_i} \nabla_{p_i} a_{ij}\left(v_2^i - v_2^j\right)^2 \end{aligned} \tag{12}$$

where $v_2 = [v_2^1, v_2^2, \cdots, v_2^N]^T$ is the eigenvector associated with λ_2 in the weighted Laplacian matrix, and $\nabla_{p_i} a_{ij}$ can be derived from (2) that

$$\nabla_{p_i} a_{ij} = -a_{ij}\left(p_i - p_j\right)/\theta^2 \tag{13}$$

Theorem 1: Consider the MRS dynamics (10), the global connectivity controller

$$u_i = \begin{cases} k_c \dfrac{\nabla_{p_i}\lambda_2}{\left\|\nabla_{p_i}\lambda_2\right\|} e^{\frac{\lambda_2 - \lambda_2^*}{\sigma}} & \text{if } \left\|\nabla_{p_i}\lambda_2\right\| \ne 0 \\ 0 & \text{otherwise} \end{cases} \tag{14}$$

can make the global connectivity converge to the interval $\left(\lambda_2^* - \varepsilon, \lambda_2^* + \varepsilon\right)$ under *Assumption 1*, where λ_2^* is the desired algebraic connectivity, and the parameter k_c and σ are defined as

$$\begin{cases} k_c > 0, \sigma < 0 & \text{if} \quad \lambda_2 < \lambda_2^* \\ k_c < 0, \sigma > 0 & \text{if} \quad \lambda_2 > \lambda_2^* \end{cases} \tag{15}$$

with $|k_c| < U_0$. In order to reduce the difficulty of online parameter tuning, we only need to change the symbols of k_c and σ without changing sizes. The steady-state error of global algebraic connectivity under the controller (14) is

$$\varepsilon = |\sigma| \ln \frac{U_0}{|k_c|} \tag{16}$$

Proof:

Consider a continuously differentiable positive function $V(\cdot):\mathbb{R}^{Nd}\to\mathbb{R}^{+}$ defined as

$$V=e^{\frac{\lambda_2-\lambda_2^*}{\sigma}} \tag{17}$$

Then, with the dynamics in (10), we can compute the time-derivative of the function as

$$\begin{aligned}\dot{V}&=\frac{\partial V}{\partial t}=\left(\frac{\partial V}{\partial p_i}\right)^T\frac{\partial p_i}{\partial t}\\&=\frac{1}{\sigma}e^{\frac{\lambda_2-\lambda_2^*}{\sigma}}\left(\nabla_{p_i}\lambda_2\right)^T\left(f_i(\boldsymbol{p})+u_i\right)\end{aligned} \tag{18}$$

From (18), by recalling that $\|\nabla_{p_i}\lambda_2\|=0$ implies $\nabla_{p_i}\lambda_2=0$, it follows that

$$\dot{V}=0 \quad \text{if } \|\nabla_{p_i}\lambda_2\|=0 \tag{19}$$

Then, let us substitute (14) into (18) and consider the case $\|\nabla_{p_i}\lambda_2\|\neq 0$.

Firstly, it is known from (15) that $k_c>0$ and $\sigma<0$ when $\lambda_2<\lambda_2^*$. Then, the function V is a decreasing function about λ_2 and

$$\begin{aligned}\dot{V}&=\frac{1}{\sigma}e^{\frac{\lambda_2-\lambda_2^*}{\sigma}}\left(\nabla_{p_i}\lambda_2\right)^T\left(f_i(\boldsymbol{p})+u_i\right)\\&\leq\frac{1}{\sigma}e^{\frac{\lambda_2-\lambda_2^*}{\sigma}}\|\nabla_{p_i}\lambda_2\|\left(-U_0+k_ce^{\frac{\lambda_2-\lambda_2^*}{\sigma}}\right)\end{aligned} \tag{20}$$

Therefore, the sufficient condition for the time-derivative $\dot{V}$ to be negative definite is that

$$-U_0+k_ce^{\frac{\lambda_2-\lambda_2^*}{\sigma}}>0 \tag{21}$$

At this point, the algebraic connectivity λ_2 increase, if

$$\lambda_2<\lambda_2^*+\sigma\, ln\frac{U_0}{k_c} \tag{22}$$

since the function V is a decreasing function about λ_2.

Similarly, it is known that $k_c<0$ and $\sigma>0$ when $\lambda_2>\lambda_2^*$. Then, the function V is an increasing function about λ_2 and

$$\begin{aligned}\dot{V}&=\frac{1}{\sigma}e^{\frac{\lambda_2-\lambda_2^*}{\sigma}}\left(\nabla_{p_i}\lambda_2\right)^T\left(f_i(\boldsymbol{p})+u_i\right)\\&\leq\frac{1}{\sigma}e^{\frac{\lambda_2-\lambda_2^*}{\sigma}}\|\nabla_{p_i}\lambda_2\|\left(U_0+k_ce^{\frac{\lambda_2-\lambda_2^*}{\sigma}}\right)\end{aligned} \tag{23}$$

The sufficient condition for the time-derivative $\dot{V}$ to be negative definite is that

$$U_0+k_ce^{\frac{\lambda_2-\lambda_2^*}{\sigma}}<0 \tag{24}$$

Therefore, the algebraic connectivity λ_2 will be reduced, if

$$\lambda_2>\lambda_2^*+\sigma\, ln\frac{U_0}{-k_c} \tag{25}$$

In summary, under the global connectivity controller proposed in (14), the connectivity will be enhanced when $\lambda_2<\lambda_2^*-\varepsilon$ and, be reduced when $\lambda_2>\lambda_2^*+\varepsilon$, so that

$$\lim_{t\to\infty}\lambda_2\in\left(\lambda_2^*-\varepsilon,\lambda_2^*+\varepsilon\right) \tag{26}$$

■

Notably, it is known from (11) that

$$\|u_i\|=|k_c|e^{\frac{\lambda_2-\lambda_2^*}{\sigma}}\leq U_c \tag{27}$$

if $\|\nabla_{p_i}\lambda_2\|\neq 0$. With (21) and (24) in the proof process, we can obtain that the upper bounds of the flocking control and the connectivity control should satisfy

$$U_0<U_c \tag{28}$$

to guarantee the convergence of the connectivity controller.

B. ANALYSIS OF FLOCKING WITH CONNECTIVITY CONTROL

It is stated in *Theorem 1* that the proposed connectivity control method can guarantee the global connectivity adjustment in the presence of the objective flocking term under the condition that $U_0<U_c$. We now investigate how the connectivity control (14) affects the achievement of the flocking of the MRS.

For the sake of generality and analysis, the objective flocking controller used in this paper is obtained by the potential-based design method, which is a popular framework widely used in the control of robotics. The main idea of the potential-based method is to define a global potential function $H(\boldsymbol{p}):\mathbb{R}^{Nd}\to\mathbb{R}$ encoding the energy of the system, for which the energy is minimized when the desired configuration is achieved [32]. Thus, the controller can be formulated as

$$F(\boldsymbol{p})=-\nabla_{\boldsymbol{p}}H(\boldsymbol{p}) \tag{29}$$

to drive robots to move towards the negative gradient of the potential function, where $F(\boldsymbol{p})$ is the stacked vector of the control term $f_i(\boldsymbol{p})$ of each robot.

Limited by the communication ability and the network topology, each robot can only obtain the energy with its neighbors. Let us define the adjacency energy of the connected robot *i* and *j* as $H_{ij}(p_i,p_j)$, and the local energy of robot *i* as $H^i(\boldsymbol{p})$. Then,

$$H^i(\boldsymbol{p})=\sum_{j\in\mathcal{N}_i}H_{ij}(p_i,p_j) \tag{30}$$

$$H(\boldsymbol{p})=\sum_{i=1}^{N}H^i(\boldsymbol{p}) \tag{31}$$

Therefore, the distributed controller is

$$f_i(\boldsymbol{p})=-\nabla_{p_i}H^i(\boldsymbol{p}) \tag{32}$$

We assume that there is an attraction between robot *i* and *j* to reduce the adjacency energy, if the adjacency distance is larger than a unique value. On the contrary, the repulsion dominates to improve the energy. Thus, the local energy can be modeled as

$$H^i(\boldsymbol{p})=H_a^i(\boldsymbol{p})+H_r^i(\boldsymbol{p}) \tag{33}$$

where $H_a^i(\boldsymbol{p})$ represents the attraction term, and $H_r^i(\boldsymbol{p})$ represents the repulsion term [33]. Correspondingly, the

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flocking control can be written as the sum of attraction and repulsion.

For the multi-robot system, there exists a unique distance r between a couple of connected robots at which the attraction and repulsion balance. At this time, the adjacency energy is minimized, and the energy gradient and control term are both zero. However, it is not conducive to the flexibility of configuration and tasks, if the desired distance between connected robots is constant. Meanwhile, the global connectivity is almost determined in this case, which is not conducive to the control of connectivity. Therefore, the adjacency potential function is defined as

$$H_{ij} = \begin{cases} \dfrac{\mu_1}{\left\|p_i - p_j\right\|^2} - \dfrac{\mu_1}{r_1^2} & if \ \left\|p_i - p_j\right\| \in (0, r_1) \\ 0 & if \ \left\|p_i - p_j\right\| \in [r_1, r_2] \\ \dfrac{\mu_2}{R^2 - \left\|p_i - p_j\right\|^2} - \dfrac{\mu_2}{R^2 - r_2^2} & if \ \left\|p_i - p_j\right\| \in (r_2, R) \end{cases} \quad (34)$$

where $\mu_1 > 0$ and $\mu_2 > 0$, and the balance distances satisfy $0 < r_1 < r_2 < R$. The adjacency potential is shown in Fig. 2. From the definition of H_{ij} in (34), we can know that the potential is balanced if and only if $\left\|p_i - p_j\right\| \in [r_1, r_2]$. Then, the gradient of adjacent potential $\nabla_{p_i} H_{ij}$ can be obtained by differentiation, and the global potential $H(\boldsymbol{p})$ can be computed with (30) and (31).

In order to fulfill *Assumption 1* and the condition in (28), the potential-based flocking controller is formulated as

$$f_i(\boldsymbol{p}) = \begin{cases} -k_0 \dfrac{\nabla_{p_i} H^i(\boldsymbol{p})}{\left\|\nabla_{p_i} H^i(\boldsymbol{p})\right\|} & if \ \left\|\nabla_{p_i} H^i(\boldsymbol{p})\right\| \neq 0 \\ 0 & otherwise \end{cases} \quad (35)$$

where k_0 is the control gain. It is known from (14) and (35) that $\|u_i\| \in [|k_c|, U_c]$ and $U_0 = k_0$. To guarantee the stability of the objective flocking controller, we cannot maintain $\left\|f_i(\boldsymbol{p})\right\| \le \|u_i\|$ all time. Thus, the parameters can be set as $|k_c| < k_0 < U_c$, where U_c is related to the maximum of the global connectivity λ_2.

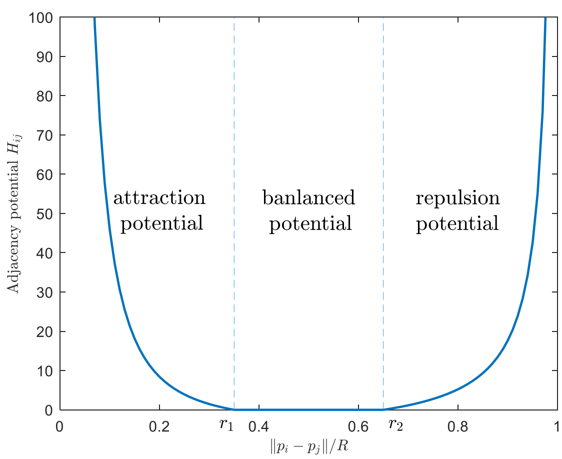

FIGURE 2. Adjacency potential curve for $r_1/R = 0.35$, $r_2/R = 0.65$

Theorem 2: Consider the MRS dynamics (10) with the global connectivity controller (14) and the objective flocking control term (35). The MRS will move towards the direction of potential decrease under the control of (35), and the energy will converge to zero.

Proof:

For the potential function satisfies $H(\boldsymbol{p}) \ge 0$, it can be utilized as the Lyapunov function to prove the stability of (35). Then,

$$\begin{aligned} \dot{H}(\boldsymbol{p}) &= \sum_{i=1}^{N} \dot{H}^i(\boldsymbol{p}) = \sum_{i=1}^{N} \left(\nabla_{p_i} H^i\right)^T \dot{p}_i \\ &= \sum_{i=1}^{N} \left(\nabla_{p_i} H^i\right)^T \left(u_i + f_i(\boldsymbol{p})\right) \end{aligned} \quad (36)$$

Since $\left\|\nabla_{p_i} H^i\right\| = 0$ implies $\nabla_{p_i} H^i = 0$, it follows that

$$\dot{H}(\boldsymbol{p}) = 0 \quad if \quad \left\|\nabla_{p_i} H^i\right\| = 0 \quad (37)$$

That is, the desired configuration is achieved and the potential will not change over time. Then, let us consider the case of $\left\|\nabla_{p_i} H^i\right\| \neq 0$.

From (35), in the case that $\left\|\nabla_{p_i} H^i\right\| \neq 0$ but $u_i = 0$ we have

$$\begin{aligned} \dot{H}(\boldsymbol{p}) &= \sum_{i=1}^{N} \left(\nabla_{p_i} H^i\right)^T \left(-k_0 \frac{\nabla_{p_i} H^i}{\left\|\nabla_{p_i} H^i\right\|}\right) \\ &= -k_0 \sum_{i=1}^{N} \left\|\nabla_{p_i} H^i\right\| < 0 \end{aligned} \quad (38)$$

It means that the global potential is reduced under the flocking control of (35) and robots move towards the desired positions.

In the case that $\left\|\nabla_{p_i} H^i\right\| \neq 0$ and $u_i \neq 0$, we have

$$\begin{aligned} &\left(\nabla_{p_i} H^i\right)^T \left(u_i + f_i(\boldsymbol{p})\right) \\ &= \left(\nabla_{p_i} H^i\right)^T \left(k_c \frac{\nabla_{p_i} \lambda_2}{\left\|\nabla_{p_i} \lambda_2\right\|} e^{\frac{\lambda_2 - \lambda_2^*}{\sigma}} - k_0 \frac{\nabla_{p_i} H^i}{\left\|\nabla_{p_i} H^i\right\|}\right) \\ &= \left\|\nabla_{p_i} H^i\right\| \left(k_c e^{\frac{\lambda_2 - \lambda_2^*}{\sigma}} \cos \beta_i - k_0\right) \end{aligned} \quad (39)$$

with β_i the angle between the two gradients $\nabla_{p_i} \lambda_2$ and $\nabla_{p_i} H^i$, that is

$$\beta_i = \angle\left(\nabla_{p_i} H^i, \nabla_{p_i} \lambda_2\right) \quad (40)$$

In this case, $k_c > 0$ and $\sigma < 0$ are defined in (15) when $\lambda_2 < \lambda_2^*$. Then, we have

$$\begin{aligned} \left(\nabla_{p_i} H^i\right)^T \left(u_i + f_i(\boldsymbol{p})\right) &= \left\|\nabla_{p_i} H^i\right\| \left(k_c e^{\frac{\lambda_2 - \lambda_2^*}{\sigma}} \cos \beta_i - k_0\right) \\ &\le \left\|\nabla_{p_i} H^i\right\| \left(k_c e^{\frac{\lambda_2 - \lambda_2^*}{\sigma}} - k_0\right) \end{aligned} \quad (41)$$

From *Theorem 1*, the global connectivity converges to the interval $\left(\lambda_2^* - \varepsilon, \lambda_2^*\right)$ under the distributed controller (14). Therefore,

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$$\left(\nabla_{p_i} H^i\right)^T \left(u_i + f_i\left(\boldsymbol{p}\right)\right) < \left\|\nabla_{p_i} H^i\right\| \left(k_c \frac{U_0}{k_c} - k_0\right) \quad (42)$$

Since $k_0 = U_0$, we have

$$\dot{H}\left(\boldsymbol{p}\right) = \sum_{i=1}^{N} \left(\nabla_{p_i} H^i\right)^T \left(u_i + f_i\left(\boldsymbol{p}\right)\right) < 0 \quad (43)$$

For the same reason, $k_c < 0$ and $\sigma > 0$ are defined if $\lambda_2 < \lambda_2^*$, and it follows that

$$\left(\nabla_{p_i} H^i\right)^T \left(u_i + f_i\left(\boldsymbol{p}\right)\right) \le \left\|\nabla_{p_i} H^i\right\| \left(-k_c e^{\frac{\lambda_2 - \lambda_2^*}{\sigma}} - k_0\right) \quad (44)$$

At this time, the connectivity converges to $\left(\lambda_2^*, \lambda_2^* + \varepsilon\right)$ under the controller (14). Then, we can obtain

$$\dot{H}\left(\boldsymbol{p}\right) = \sum_{i=1}^{N} \left(\nabla_{p_i} H^i\right)^T \left(u_i + f_i\left(\boldsymbol{p}\right)\right) < 0 \quad (45)$$

Therefore, the time derivative of the potential function is negative definite under the control of (35), in the presence of connectivity control (14).

In order to avoid the local minimizer of the potential, we consider the local potential $H^i\left(\boldsymbol{p}\right)$ defined in (30), where $H^i\left(\boldsymbol{p}\right) \ge 0$ according to (34). Then the time derivative of the local potential can be written as

$$\begin{aligned} \dot{H}^i\left(\boldsymbol{p}\right) &= \frac{\partial H^i\left(\boldsymbol{p}\right)}{\partial p_i} \cdot \frac{\partial p_i}{\partial t} = \left(\nabla_{p_i} H^i\right)^T \dot{p}_i \\ &= \left(\nabla_{p_i} H^i\right)^T \left(u_i + f_i\left(\boldsymbol{p}\right)\right) \end{aligned} \quad (46)$$

For the same reason as (37)-(45), we have $\dot{H}^i \le 0$ under the proposed method, and $\dot{H}^i = 0$ if and only if all of the adjacency potentials hold $H_{ij} = 0$. Thus, the local minimizer of the potential can be avoided. ■

According to the analysis of the connectivity control and the objective flocking control, we know that the global connectivity will firstly converge to $\left(\lambda_2^* - \varepsilon, \lambda_2^* + \varepsilon\right)$ under the controller in (14) at $U_0 < U_c$. Then, the objective flocking controller decreases the global energy to achieve the desired configuration. This means that the connectivity controller occupies a dominant role in the MRS, and the priority is given to guaranteeing the global connectivity. On this basis, the global control effect is achieved by the coordination of the two controllers.

IV. ROBUSTNESS TO ESTIMATION ERROR AND DISTURBANCE

In the previous section, we assume that the exact value of algebraic connectivity is available in the theoretical analysis of the connectivity control. However, distributed robots cannot obtain the global connectivity in the real scenario, where the algorithm in (7)-(8) is utilized to have only an estimation of λ_2. In addition, the stability of flocking configurations and the global connectivity can be destroyed due to the unknown disturbances. Thus, the robustness to the estimation error and disturbances of the proposed method is investigated in this section.

A. ROBUSTNESS TO THE ESTIMATION ERROR

In a real MRS network, the estimation error of algebraic connectivity is caused by several factors such as packet drop, time delays, position errors, and so on. Indeed, the boundedness of the estimation error is proved in [11], and the upper bound of estimation error is defined in (9). In this regard, let us replace the λ_2 in (14) by the estimation $\tilde{\lambda}_2^i$. Then we have

$$u_i = \begin{cases} k_c \dfrac{\nabla_{p_i} \tilde{\lambda}_2^i}{\left\|\nabla_{p_i} \tilde{\lambda}_2^i\right\|} e^{\frac{\tilde{\lambda}_2^i - \lambda_2^*}{\sigma}} & if \quad \left\|\nabla_{p_i} \tilde{\lambda}_2^i\right\| \ne 0 \\ 0 & otherwise \end{cases} \quad (47)$$

with the definition of parameters is the same as that of (14).

Theorem 3: Consider the MRS dynamics (10) with the connectivity control input (47) under *Assumption 1*. If the estimation error satisfies the boundedness in (9), we have

$$\lim_{t\to\infty} \lambda_2 \in \left(\lambda_2^* - \varepsilon - \varDelta, \lambda_2^* + \varepsilon + \varDelta\right) \quad (48)$$

where ε has been defined in (16).

Proof:

The proof process of this theorem is similar to that of Theorem 1. For a robot i, the Lyapunov function can be replaced as

$$\tilde{V}_i = e^{\frac{\tilde{\lambda}_2^i - \lambda_2^*}{\sigma}} \quad (49)$$

Then, the time derivative of the Lyapunov function is

$$\begin{aligned} \dot{\tilde{V}}_i &= \frac{\partial \tilde{V}_i}{\partial t} = \left(\frac{\partial \tilde{V}_i}{\partial p_i}\right)^T \frac{\partial p_i}{\partial t} \\ &= \frac{1}{\sigma} e^{\frac{\tilde{\lambda}_2^i - \lambda_2^*}{\sigma}} \left(\nabla_{p_i} \tilde{\lambda}_2^i\right)^T \left(f_i\left(\boldsymbol{p}\right) + u_i\right) \end{aligned} \quad (50)$$

It is known from the proof of Theorem 1 that the estimated global connectivity $\tilde{\lambda}_2^i$ will converge to $\left(\lambda_2^* - \varepsilon, \lambda_2^* + \varepsilon\right)$. Therefore, by exploiting (9), it follows that

$$\lim_{t\to\infty} \lambda_2 \in \left(\lambda_2^* - \varepsilon - \varDelta, \lambda_2^* + \varepsilon + \varDelta\right) \quad (51)$$

■

Thus, the existence of the estimation error $\varDelta$ has no effect on the stability of the distributed global connectivity controller, but increases the steady-state error of the control.

Theorem 4: Consider the MRS dynamics (10) with the connectivity control input (47) and the objective flocking controller (35) under *Assumption 1*. If the estimation error satisfies the boundedness in (9), the global potential function defined in (31) can converge to zero over time.

Proof:

The result follows the same reasoning as in the proof of Theorem 2, where the condition for negative definiteness of $\dot{H}\left(\boldsymbol{p}\right)$ now becomes

$$\tilde{\lambda}_2^i \in \left(\lambda_2^* - \varepsilon, \lambda_2^* + \varepsilon\right) \quad (52)$$

According to Theorem 3, the connectivity controller (47) ensures the estimated global connectivity converges to $\left(\lambda_2^* - \varepsilon, \lambda_2^* + \varepsilon\right)$ firstly. Then the system moves towards the desired configuration with the decrease of global potential, under the flocking control term in (35). ■

Therefore, the existence of the estimation error Δ has no effect on the stability and steady-state error of the objective flocking control.

B. DESIGN OF CONTROLLER TO COUNTERACT UNKNOWN DISTURBANCES

So far, the proposed methods can guarantee the desired connectivity and flocking configuration of MRS (10) with the presence of estimation error. However, in real scenarios, the dynamics of MRS follow that

$$\dot{p}_i = f_i\left(\boldsymbol{p}\right) + u_i + u_i^e \tag{53}$$

where $u_i^e \in \mathbb{R}^d$ represents the disturbance on robot i and $\left\|u_i^e\right\| \le U_e$. Therefore, we are supposed to design a robust controller to counteract the unknown disturbances.

The robust control problem has been widely investigated for decades. In recent years, an output feedback finite-time control method is addressed in [34] to stabilize the perturbed vehicle active suspension system with the existence of uncertainty or external disturbance. A combination of adaptive robust control and a terminal sliding-mode-based nonlinear disturbance observer is proposed in [35] to solve the problem of adaptive tracking control, for a class of nonlinear systems with parametric uncertainty, bounded external disturbance, and actuator saturation. Then, the effectiveness of this method is illustrated by the application to a quarter-car model with an active suspension system. As for this paper, since the main work is the analysis of the connectivity control problem, the robust controller here is only used to reject the disturbance and guarantee the stability of MRSs dynamics. Therefore, depth analysis will not be investigated, and a popular robust method based on sliding mode control is utilized in this section.

The sliding mode control is a widely used robust control method, in which the integral sliding mode can make the system states on the sliding mode manifold at the initial time and significantly increase the stability of the system. Meanwhile, the introduction of the integral term can hide the discontinuous switching and achieve continuous control, which can avoid the chattering of low-order sliding mode method [36].

Let us consider the integral sliding mode manifold

$$\delta_i = p_i\left(t\right) - p_i\left(0\right) - \int_0^t \dot{p}_i^{idl}\,dt \tag{54}$$

where $p_i\left(0\right)$ is the position of robot i at $t=0$, and $\dot{p}_i^{idl}$ represents the ideal dynamics, that is

$$\dot{p}_i^{idl} = f\left(\boldsymbol{p}\right) + u_i = k_c \frac{\nabla_{p_i}\tilde{\lambda}_2^i}{\left\|\nabla_{p_i}\tilde{\lambda}_2^i\right\|} e^{\frac{\tilde{\lambda}_2^i - \lambda_2^*}{\sigma}} - k_0 \frac{\nabla_{p_i} H^i\left(\boldsymbol{p}\right)}{\left\|\nabla_{p_i} H^i\left(\boldsymbol{p}\right)\right\|} \tag{55}$$

for the case that $\left\|\nabla_{p_i} H^i\right\| \ne 0$ and $\left\|\nabla_{p_i}\lambda_2\right\| \ne 0$, and $\dot{p}_i^{idl} = 0$ otherwise.

It is obvious that $\delta_i = 0$ at $t=0$, which is maintained over time if the ideal dynamics hold. However, the presence of disturbance u_i^e leads to $\dot{\delta}_i \ne 0$, thus the system state will deviate from the sliding manifold and $\delta_i \ne 0$ 。

Theorem 5: Consider the MRS dynamics (53) with the connectivity control input (47) and the flocking controller (35), the additional robust control term

$$\begin{cases} u_i^c = -k_P\delta_i + \varphi_i \\ \dot{\varphi}_i = -k_I\delta_i \end{cases} \tag{56}$$

can maintain system states on the sliding mode manifold and guarantee the stability of controller (35) and (47) under the disturbance of u_i^e, where $k_P > 0$ and $k_I > 0$ are control parameters.

Proof:

According to Theorem 5, the dynamics of the MRS can be rewritten as

$$\dot{p}_i = f_i\left(\boldsymbol{p}\right) + u_i + u_i^c + u_i^e \tag{57}$$

Substituting (57) into the time derivative of the sliding mode vector δ_i, we have

$$\begin{aligned} \dot{\delta}_i &= \dot{p}_i - \left(f\left(\boldsymbol{p}\right) + u_i\right) \\ &= \left(f\left(\boldsymbol{p}\right) + u_i + u_i^c + u_i^e\right) - \left(f\left(\boldsymbol{p}\right) + u_i\right) \\ &= u_i^c + u_i^e \end{aligned} \tag{58}$$

In the case that $\dot{\delta}_i \ne 0$ and $\delta_i \ne 0$ under the effect of the disturbance u_i^e, the Laplace transform of (58) and δ_i can be written as

$$s\mathcal{L}\left(\delta_i\right) = -k_P\mathcal{L}\left(\delta_i\right) - \frac{1}{s}k_I\mathcal{L}\left(\delta_i\right) + \mathcal{L}\left(u_i^e\right) \tag{59}$$

$$\mathcal{L}\left(\delta_i\right) = \frac{s\mathcal{L}\left(u_i^e\right)}{s^2 + k_P s + k_I} \tag{60}$$

where $\mathcal{L}(\cdot)$ is the Laplace transform function, and s is a complex variable. Since $k_I > 0$ and $\left\|u_i^e\right\| \le U_e$, according to the final value theorem, we have

$$\lim_{t\to\infty}\delta_i = \lim_{s\to 0} s\mathcal{L}\left(\delta_i\right) = \lim_{s\to 0}\frac{s^2\mathcal{L}\left(u_i^e\right)}{s^2 + k_P s + k_I} = 0 \tag{61}$$

$$\lim_{t\to\infty}\dot{\delta}_i = \lim_{s\to 0} s^2\mathcal{L}\left(\delta_i\right) = 0 \tag{62}$$

Therefore, the system states can converge to the sliding mode manifold under the controller (56) if $\delta_i \ne 0$.

Let us consider the case $\delta_i = \dot{\delta}_i = 0$. It is known from (58) that $u_i^c + u_i^e = 0$, which means the integral sliding mode controller can completely counteract the disturbance of the system and make the system the ideal dynamics as (10). In addition, the control strategies of (10) have been proved in this paper.

Thus, the unknown disturbances can be counteracted by the proposed robust controller in (56) which has no effect on the stability of the original controllers. ■

V. SIMULATIONS

In this section, simulations have been implemented in MATLAB to validate the proposed control methods. First, from the comparison of the global algebraic connectivity and the distributed connectivity estimation, we analyze the

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tracking effect and the estimation error of the estimation method. Then, simulations are carried out to illustrate the proposed method in this paper.

TABLE I
PARAMETERS SETTING FOR CONNECTIVITY ESTIMATION

Symbol	k_1	k_2	k_3	k_4	γ	η_1	η_2
Value	18	3	60	0.01	100	50	200

We consider a MRS that consists of $N=7$ robots, and the communication radius of each robot is defined as $R=100m$. Then, an initial configuration can be generated in a $200m\times 200m\times 200m$ space randomly.

TABLE I lists the parameters setting used in the estimation of global connectivity. Indeed, these parameters are independent of controllers and can be set in advance without the online adjustment.

In order to validate the robustness of the proposed control method, we consider the dynamics in (57), that is

$$\dot{p}_i = f_i\left(\boldsymbol{p}\right)+u_i+u_i^c+u_i^e$$

where the disturbance of the system is defined as

$$u_i^e = k\left[sin\left(\frac{2\pi}{N}i\right) \quad cos\left(\frac{2\pi}{N}i\right) \quad sin\left(\frac{\pi}{N}i\right)\right]^T \tag{63}$$

with $k>0$ the coefficient of disturbance and now $k=2$.

In addition, to avoid the oscillation caused by the symbol switching of parameters in the connectivity controller (47), we consider

$$\begin{aligned} k_c &= k_{c0}sat\left(\lambda_2^*-\tilde{\lambda}_2^i\right) \\ \sigma &= \sigma_0 sat\left(\tilde{\lambda}_2^i-\lambda_2^*\right) \end{aligned} \tag{64}$$

where $k_{c0}=1,\sigma_0=1$, and the function $sat(\cdot)$ is defined as

$$sat\left(x\right)=\begin{cases} 1 & , \quad x>\xi \\ \dfrac{x}{\xi} & , \quad \left|x\right|\le\xi \\ -1 & , \quad x<-\xi \end{cases} \tag{65}$$

with $\xi=0.2$.

Note that, the too-large distance between robots will lead to a significant reduction in communication quality, while too little distance will increase the risk of collision. Thus, the desired maximum and minimum distance of MRS flocking is set to $r_1=30m$ and $r_2=70m$, respectively. TABLE II summarizes the parameters setting of proposed controllers.

TABLE II
PARAMETERS SETTING FOR PROPOSED CONTROLLERS

Symbol	k_0	μ_1	μ_2	k_P	k_I
Value	1.2	1	10	5	1

The comparison of the global algebraic connectivity and the distributed connectivity estimation is shown in Fig. 3, where the global connectivity is $\lambda_2\left(0\right)=0.5$ at the initial time. The global connectivity is expected to increase to the desired $\lambda_2^*=3$ and then decrease to $\lambda_2^*=1.5$. It is shown from the figure that the distributed connectivity estimator can track and estimate the global connectivity dynamically in the whole process. Meanwhile, the estimation error is bounded in this scenario and is on the order of 10^{-2}. Based on the effective estimation of the global connectivity, let us carry out simulations to analyze the ability of the proposed control methods.

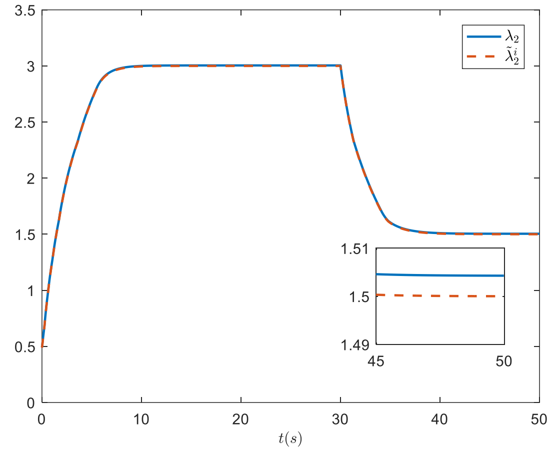

FIGURE 3. Comparison of the global algebraic connectivity and the distributed connectivity estimation with *i*=1

Fig. 4 depicts the difference of a MRS running the global connectivity control (47) and flocking control (35) with and without the robust controller u_i^c. In this scenario, the MRS is supposed to maintain the global connectivity that $\lambda_2^*=\lambda_2\left(0\right)=1$. Fig. 4(a) depicts the initial configuration of the MRS, while Fig. 4(b) illustrates the final configuration with the robust controller u_i^c, and Fig. 4(c) shows the final configuration without the robust controller. It can be seen in Fig. 4(c) that robots move towards two directions under the effect of disturbances, where some of the initial links are interrupted due to the increase of the adjacency distances while some robots face the great risk of collision for too small distances. The final configuration in Fig. 4(b) and the sliding mode parameters in Fig. 4(d) indicate that the proposed robust controller (56) can counteract the influence of the disturbance to achieve the state $\delta_i=\dot{\delta}_i=0$. Thus, we obtain the ideal dynamics in (10). Fig. 4(e) illustrates the change of the global connectivity. It is noticed that the global connectivity decreases gradually and the network has the trend of disconnection under the effect of the disturbance, which is consistent with the configuration in Fig. 4(c). With the addition of the robust controller (56), the global connectivity is stable at $\lambda_2^*+\varepsilon+\varDelta$, which is consistent with the theoretical analysis in Theorem 3. The change of global potential in Fig. 4(f) follows the same reason as the connectivity. However, it is known from the definition of the potential function that the adjacency potential will drop to zero, if the adjacency distance increases to $\left\|p_i-p_j\right\|>R$. Therefore, the increase process of the global potential in Fig. 4(f) is accompanied by several times of potential decrease.

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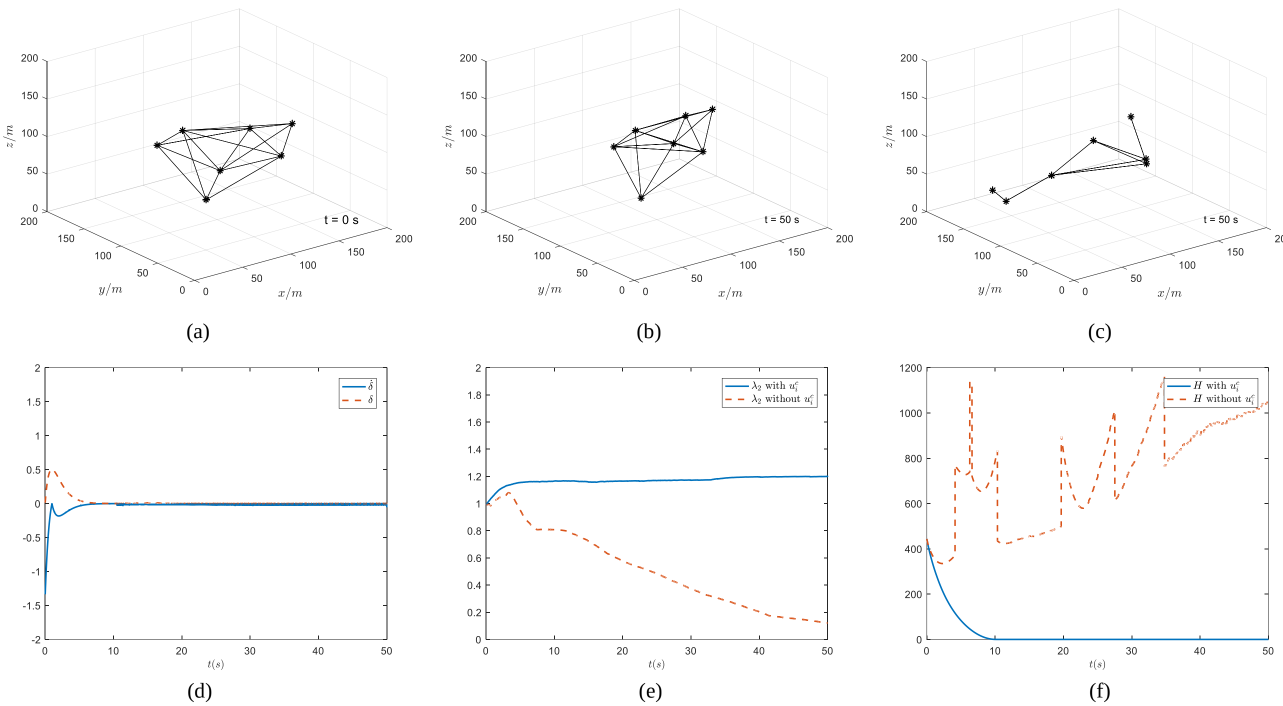

FIGURE 4. MRS running the connectivity and flocking control with the robust controller compared with the case without the robust controller

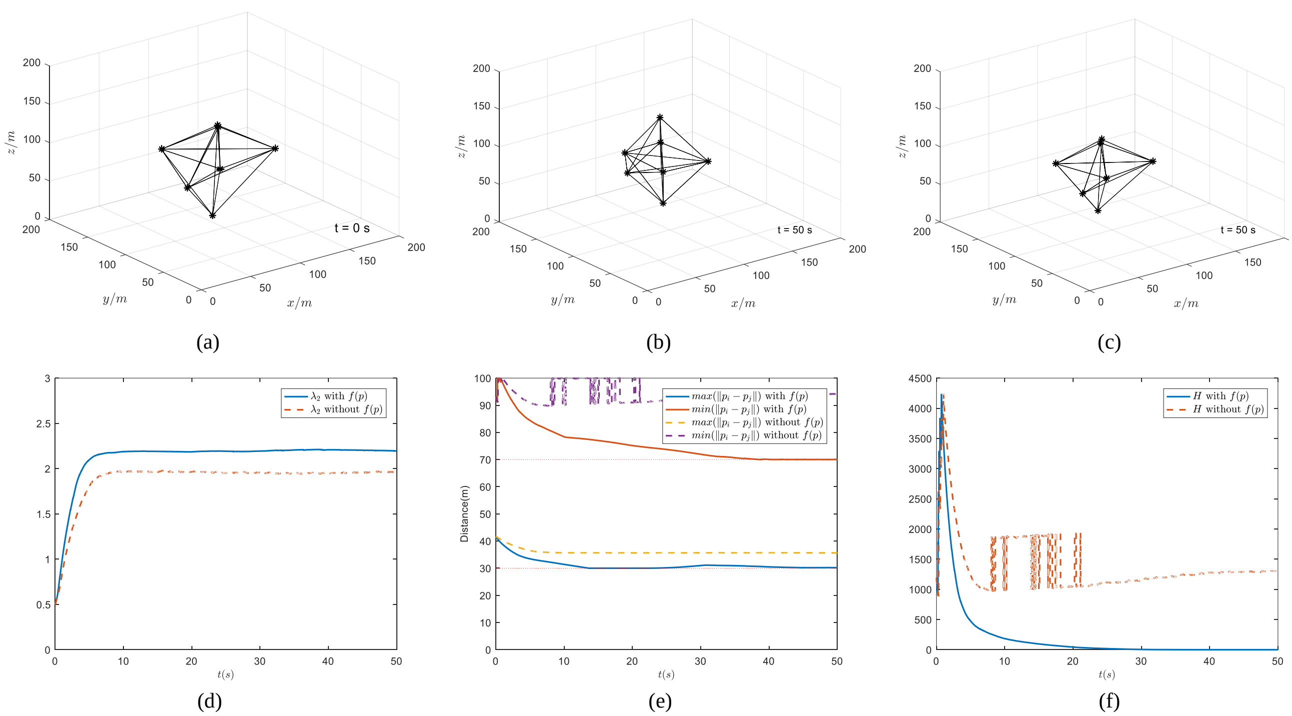

FIGURE 5. MRS running the connectivity and robust control with the flocking controller compared with the case without flocking controller

Fig. 5 depicts the behavior of a MRS running the global connectivity control (47) and robust control (56) with and without the objective flocking controller $f(\boldsymbol{p})$. In this scenario, the MRS is expected to improve the global connectivity from $\lambda_2(0)=0.5$ to the desired $\lambda_2^*=2$. Fig. 5(a) depicts the initial configuration of the MRS, while Fig. 5(b) shows the final configuration with the objective flocking controller $f(\boldsymbol{p})$, and Fig. 5(c) illustrates the final configuration without the flocking controller. It can be noticed that the configuration is more uniform under the flocking control in (35), while the final configuration maintains the initial shape without the flocking control, where only adjacency distances are shrunk to increase the global connectivity. Fig. 5(d) shows how the algebraic connectivity changes over time. It can be seen from the figure that the algebraic connectivity converges to

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$\lambda_2^* + \varepsilon + \varDelta$ with the addition of $f(\boldsymbol{p})$, while it converges to $\lambda_2^* + \varDelta$ without $f(\boldsymbol{p})$. This result corresponds to the proof process of Theorem 3, that is, we have $\dot{V} < 0$ for $\forall \lambda_2 \neq \lambda_2^*$ in the case of $U_0 = 0$, while the steady-state error ε exists if $U_0 > 0$. Since we have assumed the presence of $f(\boldsymbol{p})$, the case of $U_0 = 0$ was not discussed in the proof process of Theorem 3. Fig. 5(e) depicts the change of maximum and minimum adjacency distances, and Fig. 5(f) shows the change of global potential. It can be seen from Fig. 5(e) that the maximum adjacency distance of the system is always at a large value when only the connectivity control is utilized. Meanwhile, during the time of 10-20s, the states of some communication links switch between "connected" and "disconnected" frequently since the adjacency distance close to communication radius, which leads to extremely poor communication quality and consumes a lot of resources in practical applications. This situation is also reflected in the potential curve of Fig. 5(f). By adding the flocking control of (35), the global adjacency distances can be stabilized within the desired range, which can not only ensure the stability of the communication link but also avoid collision of neighbors. In general, the flocking control has no effect on the stability of the proposed connectivity control, but ensures the quality of communication links and the security of MRS motion at the expense of connectivity control accuracy.

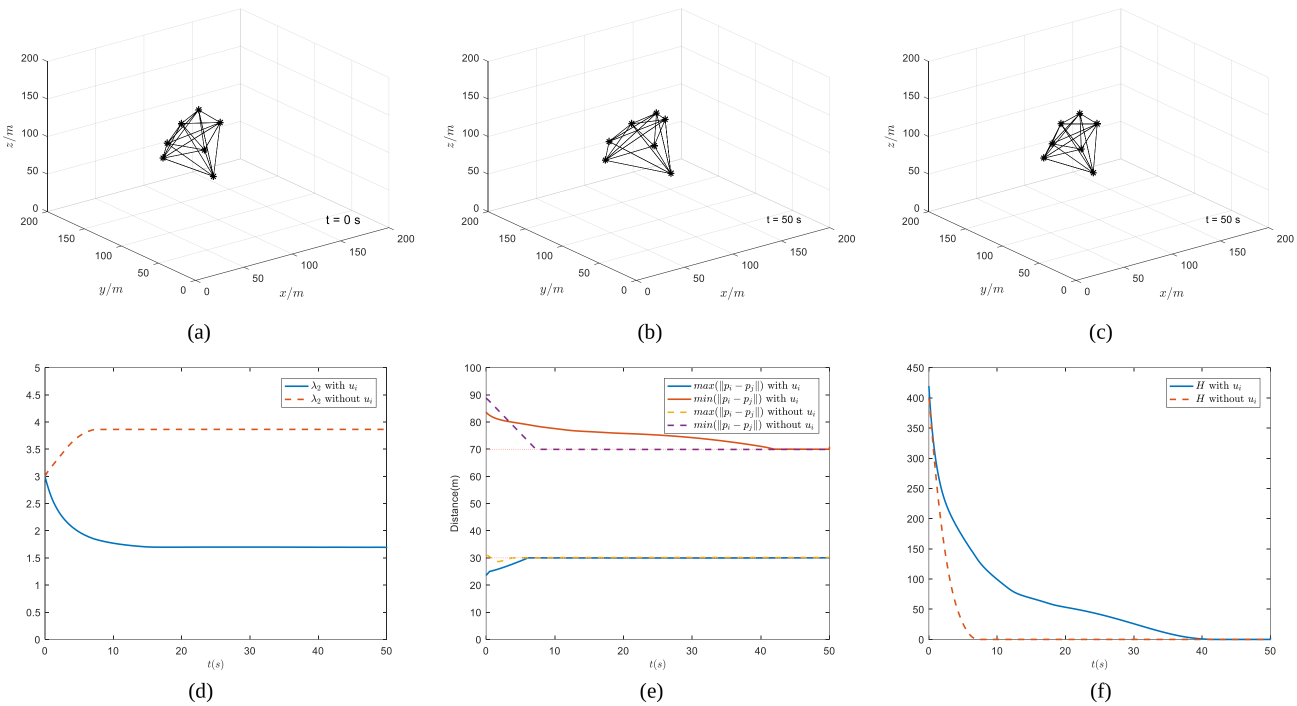

FIGURE 6. MRS running the flocking and robust control with the connectivity controller compared with the case without connectivity controller

Fig. 6 illustrates the difference of a MRS running the global flocking control (35) and robust control (56) with and without the connectivity controller u_i. In this scenario, the MRS is supposed to improve the global connectivity from $\lambda_2(0) = 3$ to the desired $\lambda_2^* = 1.5$. Fig. 6(a) depicts the initial configuration of the MRS, while Fig. 6(b) illustrates the final configuration with the connectivity controller u_i, and Fig. 6(c) shows the final configuration without the connectivity controller. It is shown from the figures that the configuration is well maintained under both conditions, and the configuration is more compact without the connectivity control. Fig. 6(d) depicts the change of the algebraic connectivity over time. It can be noticed that the algebraic connectivity approaches the value $\lambda_2^* + \varepsilon + \varDelta$ with the addition of (47). However, in the absence of u_i, the algebraic connectivity is much greater than the expected value, which is also consistent with the more compact configuration in Fig. 6(c). Fig. 6(e) illustrates the maximum and minimum adjacency distances in the MRS, and Fig. 6(f) shows how the global potential changes over time. It can be seen from Fig. 6(e) that the maximum adjacency distance of the system can quickly converge to the desired range if the flocking control is utilized only. After the addition of the connectivity control u_i, the maximum adjacency distance can also converge to the expected range, but it is slower than the former case. The global potential in Fig. 6(f) follows the same changes as adjacency distances. Therefore, the connectivity control has no effect on the stability of the flocking control, but guarantees the desired global algebraic connectivity at the expense of the convergence time of configuration.

VI. CONCLUSION

In this paper, we investigated the connectivity control problem for multi-robot systems with an objective flocking control term. We proposed a gradient-based control method to adjust the global connectivity according to task situations.

Meanwhile, we addressed the effect between the proposed connectivity control and the objective potential-based flocking control. We also analyzed the robustness of the connectivity control method to the estimation error of the algebraic connectivity, and designed a robust method based on integral sliding mode to counteract the disturbances of the system. The effectiveness of the proposed methods was validated in simulations. In future work, we will consider more complex communication models including time delay, packet drop and communication interruption to make the connectivity control method more practical. Meanwhile, more advanced robust design methods will be discussed to enhance the robustness of connectivity control of multi-robot systems.

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